

OceanEmbed

Satellite-to-Subsurface Ocean Temperature Reconstruction

SIH 2026 Technical Understanding Dossier & Jury Master Reference | North Indian Ocean Domain

EXECUTIVE SUMMARY & OPERATIONAL CONTEXT

The Operational Challenge: Satellites provide continuous, basin-wide monitoring of surface ocean properties (SST, SSS, SSH, wind velocity, surface currents). However, vital oceanographic phenomena—including the **main thermocline**, upper-ocean heat content (OHC), tropical cyclone heat potential, and underwater acoustic propagation—are governed by the **subsurface vertical temperature structure down to 1000 meters**.

In-situ observation systems (such as ARGO profiling floats, CTD casts, and moored buoys) provide high vertical accuracy but are spatially sparse and temporally intermittent. **OceanEmbed** solves this fundamental observational gap by establishing a deep-learning framework that learns the non-linear physical mapping from surface satellite observation patterns to the complete 15-depth vertical temperature column on a continuous daily 0.25° grid across the North Indian Ocean.

END-TO-END RECONSTRUCTION PIPELINE

1. SURFACE INPUTS	2. SPATIAL ENCODER	3. OCEAN EMBEDDING	4. MULTI-DEPTH HEAD
7 Physical Surface Observations 7 Binary Validity Quality Masks - Grid: 101 × 241 (0.25° res) 14 Channels Total	- Hierarchical U-Net DoubleConv - Multi-Scale Spatial Pooling - Skip Feature Routing - Receptive Field: Basin-wide	128-d Latent Bottleneck - Coupled Cross-Variable Features - Decoder Transpose-Conv - Dimension: [128, 12, 30]	1×1 Parallel Projection Head - Learned Stratification Prior 15 Depths Simultaneously 0m to 1000m Output Column
Target Domain 5°N–30°N, 45°E–105°E (Arabian Sea + Bay of Bengal)	Spatial Grid 0.25° × 0.25° Discretization 101 Lats × 241 Lons	Vertical Column 15 Locked Depths 0m to 1000m Strata	Cadence & Archive Daily Snapshots 274 Days (Jan–Sep 2020)

VERIFIED SCIENTIFIC & COMPUTATIONAL BENCHMARKS

0.8601°C GLORYS Reference RMSE -17.4% vs Simple CNN	0.7973°C ARGO Observational RMSE -16.3% vs Simple CNN	7.2× Training Speedup Parallel Multi-Depth Head	~35.9 ms Mean API Latency CPU Benchmark (Local)
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SCIENTIFIC DISCIPLINE & LEAKAGE PREVENTION

Rigorous Scientific Guardrails: The OceanEmbed framework enforces strict temporal splitting (Train: Jan–Jul 2020 | Validation: Aug 2020 | Held-out Test: Sep 2020). Normalization scalars are fit strictly on training samples. In-situ ARGO observations are **never used during training or architecture selection**, serving exclusively as an independent blind observational check. Numerical reanalysis (GLORYS) is treated as a physical reference target rather than a direct in-situ sensor measurement.

Data Architecture & Problem Formulation

PROBLEM FORMULATION & OBJECTIVE FUNCTION

Mathematical Formulation: Let X_t in $\mathbb{R}^{14 \times H \times W}$ denote the multi-modal gridded surface ocean state at calendar day t , comprising 7 standardized physical observations and 7 binary validity masks over a regular 0.25° grid ($H = 101, W = 241$). The objective is to estimate the vertical temperature field Y_t in $\mathbb{R}^{15 \times H \times W}$ across 15 standard subsurface depths: $D = \{0, 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 300, 500, 700, 1000\}$ m.

The learning task minimizes the masked mean squared error over all valid ocean pixels: $\mathcal{L}(\theta) = || M_{\text{ocean}} * (Y - f_{\theta}(X)) ||^2 / || M_{\text{ocean}} ||_1$, where M_{ocean} masks out land and bathymetric barriers.

SURFACE INPUT DATA CATALOGUE & PROVENANCE

Channel	Physical Variable	Provider / Product Identifier	Native Res	Temporal	Physical Units
Ch 0 (Mask Ch 7)	Sea Surface Temperature (SST)	UK Met Office / OSTIA L4 REP	0.05°	Daily	Celsius (°C)
Ch 1 (Mask Ch 8)	Sea Surface Salinity (SSS)	Copernicus Multi-Obs (SMOS/SMAP)	0.25°	Weekly -> Daily	PSU
Ch 2 (Mask Ch 9)	Sea Surface Height (SSH/SLA)	Copernicus DUACS Altimetry L4	0.125°	Daily	Meters (m)
Ch 3 (Mask Ch 10)	Surface Zonal Current (U)	Copernicus Multi-Obs (Altimetry+Drifters)	0.25°	Daily	m/s
Ch 4 (Mask Ch 11)	Surface Meridional Current (V)	Copernicus Multi-Obs (Altimetry+Drifters)	0.25°	Daily	m/s
Ch 5 (Mask Ch 12)	10m Zonal Neutral Wind (U)	Copernicus Scatterometer L4	0.125°	Hourly -> Daily	m/s
Ch 6 (Mask Ch 13)	10m Meridional Wind (V)	Copernicus Scatterometer L4	0.125°	Hourly -> Daily	m/s

BINARY MASKING & QUALITY CONTROL PROTOCOL

Quality Control & NaN Propagation Prevention: Satellite sensors frequently encounter cloud cover, landmasses, or shallow bathymetric cutoffs. If unhandled, convolutional filters spread NaNs across valid marine cells. OceanEmbed handles this via a dual-channel strategy: in physical channels 0–6, invalid/missing cells are set to **0.0**, while mask channels 7–13 supply explicit binary indicators (**1.0 = authentic ocean observation, 0.0 = missing/land**). The network's first convolution learns to weight inputs proportionally to their mask validity.

3D SUBSURFACE REFERENCE TARGET SPECIFICATION (GLORYS)

Target Reference Dataset GLORYS12V1 Global Reanalysis Copernicus GLOBAL_MULTIYEAR_PHY_001_030	Target Physical Variable 3D Potential Temperature (thetao) Units: Degrees Celsius (°C)	Vertical Interpolation 1D Linear Interpolation across 50 native levels Bounded [0.494m, 1062.5m], Surface at 0m	Horizontal Alignment Bilinear regridding $0.083^\circ \rightarrow 0.25^\circ$ Strict retention of bathymetric NaNs
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DATA PIPELINE & CHUNKING SUMMARY

Assembly Pipeline: (1) Daily netCDF extraction -> (2) Horizontal regridding to 0.25° grid -> (3) Vertical interpolation of GLORYS thetao to 15 depths -> (4) 14-channel tensor assembly with binary masks -> (5) Storage in monthly chunk archives (chunk_2020_01.pt to chunk_2020_09.pt). **Audit Verification:** All 274 daily fields (Jan 1 – Sep 30, 2020) verified present with 0 duplicate and 0 missing dates.

Deep Neural Architecture: OceanEmbedNetV3_Decoder

STRUCTURAL BREAKTHROUGH: PARALLEL MULTI-DEPTH PROJECTION

Architectural Evolution & Bottleneck Elimination: The baseline Phase 4 model (OceanEmbedNet) suffered from a severe computational bottleneck: it passed a scalar depth indicator through a shared decoder, sequentially evaluating the 15 depth layers one by one in an explicit loop. This caused long training times (~172.4 s/epoch) and inter-depth interference.

OceanEmbedNetV3_Decoder resolves this by replacing sequential decoding with a **parallel multi-depth vertical projection head**. A hierarchical U-Net encoder extracts coupled multi-scale surface features, compresses them into a 128-dimensional latent ocean embedding, decodes them back to spatial resolution [B, 32, 101, 241], and projects directly to all 15 depths simultaneously via a single 1x1 convolution. This yielded a **7.2x training throughput speedup (~23.9 s/epoch)** under identical hardware.

STAGE 01: Multi-Scale U-Net Encoder

Input Tensor: [B, 14, 101, 241] (7 Surface Variables + 7 Quality Masks)

- **enc1:** DoubleConv(14 -> 32) -> Output: [B, 32, 101, 241] | MaxPool2d(2) -> [B, 32, 50, 120]
- **enc2:** DoubleConv(32 -> 64) -> Output: [B, 64, 50, 120] | MaxPool2d(2) -> [B, 64, 25, 60]
- **enc3:** DoubleConv(64 -> 128) -> Output: [B, 128, 25, 60] | MaxPool2d(2) -> [B, 128, 12, 30]

STAGE 02: Latent Ocean Embedding Bottleneck

Bottleneck Tensor: [B, 128, 12, 30] (128-channel compact latent representation)

- Captures non-linear cross-variable dependencies between SST, SSS, SSH, currents, and atmospheric wind fields.
- Encodes spatial geostrophic and thermal equilibrium patterns across the North Indian Ocean basin.

STAGE 03: Spatial Decoder with Lateral Skip Connections

Upsampling & Multi-Scale Feature Routing:

- **up3 + dec3:** ConvTranspose2d(128 -> 128) + Concat(enc3, 128) -> DoubleConv(256 -> 128) -> [B, 128, 25, 60]
- **up2 + dec2:** ConvTranspose2d(128 -> 64) + Concat(enc2, 64) -> DoubleConv(128 -> 64) -> [B, 64, 50, 120]
- **up1 + dec1:** ConvTranspose2d(64 -> 32) + Concat(enc1, 32) -> DoubleConv(64 -> 32) -> [B, 32, 101, 241]

STAGE 04: Dedicated Multi-Depth Projection Head & Stratification Prior

Simultaneous 15-Depth Reconstruction:

- **final_conv:** nn.Conv2d(32, 15, kernel_size=1) directly maps full-resolution features to 15 depths.
- **climatology_prior:** Learnable parameter vector [1, 15, 1, 1] initialized with Indian Ocean baseline stratification.
- **Output Field:** [B, 15, 101, 241] temperature predictions across all 15 depths simultaneously.

Total Trainable Parameters 1,275,934 Weights	Total State Dict Elements 1,278,252 (incl. 2,318 BatchNorm buffers)	Receptive Field Global Basin Coverage	Checkpoint SHA-256 29E94F9CC28B13...351CF2
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INDUCTIVE BIAS: CLIMATOLOGICAL VERTICAL PRIOR

Physical Inductive Bias (Climatological Prior): Tropical ocean basins exhibit strong stable vertical thermal stratification (warm surface mixed layer decaying monotonically into intermediate and cold deep waters). OceanEmbedNetV3_Decoder incorporates a learnable prior tensor initialized with regional North Indian Ocean mean temperatures [28.2, 28.1, 27.9, 27.5, 26.8, 25.0, 22.5, 19.8, 17.2, 15.1, 13.0, 10.5, 8.4, 7.0, 5.2] °C. The final projection layer's bias is initialized to zero, ensuring the network begins training by predicting anomaly offsets relative to the physical stratification curve.

Experimental Validation & Model Selection

STRICT TEMPORAL SPLITTING (ZERO FUTURE-PERIOD CONTAMINATION)

TRAINING DATASET January 1 – July 31, 2020 213 Daily Fields Model weight optimization	VALIDATION DATASET August 1 – August 31, 2020 31 Daily Fields Model Selection Criterion	HELD-OUT TEST DATASET September 1 – September 30, 2020 30 Daily Fields Final Benchmark Evaluation
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MODEL SELECTION INTEGRITY (AUGUST 2020 GATE)

Model Selection Evidence: In Phase 5, all candidate architectures were trained strictly on Jan–Jul 2020 data and compared on the **August 2020 validation set**. OceanEmbed v3 (EXP-02) achieved the lowest validation error: **0.7570°C RMSE** (vs Original OceanEmbed: 1.8867°C, Pointwise MLP: 1.0836°C, Simple CNN: 0.9739°C, EXP-01: 1.8978°C, EXP-04: 0.8433°C). **Critical Scientific Proof:** September GLORYS reference data and September ARGO observational data were completely held out and **never accessed during model selection or hyperparameter tuning**.

HELD-OUT SEPTEMBER 2020 GLORYS TEST RESULTS (4,601,790 VALID OCEAN CELLS)

Model Architecture	Parameters	GLORYS RMSE	GLORYS MAE	Pearson Corr (r)	vs Simple CNN
Static Regional Climatology	15 (prior)	2.9976 °C	2.4705 °C	0.9663	Baseline Reference
Pointwise MLP Baseline	34, 831	1.1177 °C	0.7569 °C	0.9893	+7.3% error
Simple CNN Baseline	1, 210, 639	1.0418 °C	0.7100 °C	0.9907	Standard Baseline
Original OceanEmbedNet (Phase 4)	1, 342, 928	1.8142 °C	1.3229 °C	0.9741	+74.1% error
OceanEmbedNetV3_Decoder (Ours)	1, 275, 934	0.8601 °C	0.5780 °C	0.9936	-17.44% error

INDEPENDENT OBSERVATIONAL CHECK (ARGO FLOATS)

ARGO Observational Verification Protocol - Independent GDAC in-situ datasets: September 1, 15, 25, 2020. 36 authentic ARGO profiles across 28 unique WMO floats. 497 matched profile-depth observation pairs (14 depths, 5m to 1000m). - Uncalibrated dummy-zero profiles strictly filtered. - Exact calendar-day temporal matching and spatial nearest-centroid collocation. - Note: Surface 0m has no ARGO sensor measurements.	In-Situ Observational Results - Static Climatology: 3.0363 °C RMSE - Pointwise MLP: 1.1602 °C RMSE - Original OceanEmbedNet: 1.8126 °C RMSE - Simple CNN Baseline: 0.9526 °C RMSE OceanEmbed v3 (Ours): 0.7973 °C RMSE Relative Gain: 16.30% error reduction vs CNN
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SCIENTIFIC CLAIM DISCIPLINE

Rigorous Verification Nomenclature: The validation sample comprises exactly **497 matched profile-depth observations from 36 ARGO profiles across 28 unique WMO floats** across 14 evaluated depth levels. ARGO profiles serve as an in-situ observational check rather than an absolute reference standard.

Depth-Wise Verification & Live System Integration

DEPTH-WISE ARGO RMSE EVALUATION (OCEANEMBED V3 OUTPERFORMS CNN AT ALL 14 DEPTHS)

Depth Level	Ocean Regime	Simple CNN RMSE	OceanEmbed v3 RMSE	Delta Error	Relative Gain
5 m	Near-Surface	0.5825 °C	0.5284 °C	-0.0541 °C	+9.3%
10 m	Mixed Layer	0.7618 °C	0.6030 °C	-0.1588 °C	+20.8%
20 m	Mixed Layer	0.8695 °C	0.6582 °C	-0.2113 °C	+24.3%
30 m	Mixed Layer	0.9604 °C	0.7194 °C	-0.2410 °C	+25.1%
50 m	Upper Thermocline	1.0366 °C	0.8679 °C	-0.1687 °C	+16.3%
75 m	Main Thermocline	1.2220 °C	1.0388 °C	-0.1832 °C	+15.0%
100 m	Peak Thermocline	1.6876 °C	1.5317 °C	-0.1559 °C	+9.2%
125 m	Main Thermocline	1.1984 °C	1.0631 °C	-0.1353 °C	+11.3%
150 m	Lower Thermocline	1.0278 °C	0.8645 °C	-0.1633 °C	+15.9%
200 m	Intermediate	1.1067 °C	0.9100 °C	-0.1967 °C	+17.8%
300 m	Intermediate	0.6147 °C	0.3585 °C	-0.2562 °C	+41.7%
500 m	Deep Ocean	0.4532 °C	0.2997 °C	-0.1535 °C	+33.9%
700 m	Deep Ocean	0.3757 °C	0.2631 °C	-0.1126 °C	+30.0%
1000 m	Deep Ocean	0.4037 °C	0.2586 °C	-0.1451 °C	+35.9%

REGIME ANALYSIS: THERMOCLINE DYNAMICS & DEEP STABILITY

Thermocline & Deep Ocean Insights: (1) The main thermocline (75–150m) exhibits the largest absolute errors across all models due to intense vertical temperature gradients (>0.1°C/m) and internal wave displacement. OceanEmbed v3 achieved consistent error reductions of 0.14°C to 0.18°C across this zone. (2) In the deep ocean (500–1000m), OceanEmbed v3 maintained exceptional physical stability, reaching **0.2586°C RMSE at 1000m** (a 35.9% improvement over the baseline CNN's 0.4037°C).

REAL LIVE INFERENCE DEMONSTRATION (CENTRAL BAY OF BENGAL)

Demonstration Coordinate: 15.00°N, 85.00°E (Central Bay of Bengal) Date: 2020-09-15 Extracted Surface Observations: • SST: 29.547 °C SSS: 33.397 psu SSH: 0.200 m • Zonal Current (U): -0.051 m/s Meridional Current (V): 0.168 m/s • Zonal Wind (U): 5.017 m/s Meridional Wind (V): 6.696 m/s • Ocean Water Validity: True (Mask = 1.0)	Reconstructed 15-Depth Vertical Column: • 0m: 29.65°C 5m: 29.59°C 10m: 29.56°C • 20m: 29.61°C 30m: 29.53°C 50m: 28.94°C • 75m: 26.82°C 100m: 23.75°C 125m: 20.60°C • 150m: 17.92°C 200m: 14.68°C 300m: 11.90°C • 500m: 10.04°C 700m: 8.58°C 1000m: 6.68°C Derived Indicators: MLD = 32.88 m Thermocline = 112.5 m OHC300 = 24.465 GJ/m²
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INFERENCE ENGINE BENCHMARK (LOCAL CPU PROFILING)

Backend Architecture & Latency Benchmark: Built in FastAPI with a singleton service architecture (model weights and scaler loaded once on startup). Pure neural model forward pass requires **31.85 ms**. Total roundtrip API latency (including coordinate mapping, scaler normalization, and JSON serialization) averages **35.87 ms (median: 34.92 ms, p95: 44.84 ms)** on standard host CPU, supporting interactive real-time dashboard updates.

Demonstration, Scientific Boundaries & Jury Defense

END-TO-END DEMO EXECUTION ARCHITECTURE

Interactive Live Demonstration Workflow:

- User Interaction:** The operator selects coordinates on the interactive Leaflet map or enters custom latitude/longitude within 5°N–30°N, 45°E–105°E and chooses a snapshot date from the 2020 archive (274 available dates).
- API Dispatch:** The React/Vite frontend dispatches a JSON payload to the FastAPI endpoint POST /predict.
- Zero-Leakage Preprocessing:** The backend validates coordinates, rejects land pixels (via Channel 7 mask), extracts the 14-channel input field, and normalizes using the frozen training scaler.
- Neural Forward Pass:** OceanEmbed v3 predicts the 15-depth temperature profile in ~35 ms under torch.inference_mode().
- Diagnostics & SVG Rendering:** The frontend dynamically plots the vertical temperature curve, calculates Mixed Layer Depth (MLD), locates the thermocline inflection depth, and computes Upper Ocean Heat Content (OHC300).

SCIENTIFIC BOUNDARIES & EXPLICIT PROTOTYPE LIMITATIONS

Operational Boundary	Documented Scope & Scientific Qualification
Archive Demonstration	Operates on a reprocessed 2020 archive (274 daily fields). This is a research prototype, not a real-time operational satellite feed.
Reanalysis Target	GLORYS12V1 is used as a numerical reanalysis reference target, rather than a direct in-situ observational sensor measurement.
Observational Sample	ARGO verification is based on selected September 2020 snapshots (497 matched profile-depth pairs across 28 floats).
Temporal Generalization	Extending reconstruction across different years and extreme climate modes (e.g. positive/negative IOD, El Niño) remains future validation scope.
Domain Restriction	Valid predictions are strictly constrained to the North Indian Ocean basin (5°N–30°N, 45°E–105°E) down to 1000m depth.

ANTICIPATED JURY DEFENSE & TECHNICAL ANSWERS

Anticipated Jury Defense Guide:

- Q1: Did you train on ARGO or select models using ARGO?**

Answer: Absolutely not. ARGO observations were completely untouched throughout model development and hyperparameter tuning. OceanEmbed v3 was selected based strictly on August 2020 GLORYS validation. ARGO was used solely as a blind observational reality check.

- Q2: Why does error peak around 100m depth?**

Answer: The 75–150m layer represents the tropical permanent thermocline, where vertical temperature gradients exceed 0.1°C/m and internal waves induce high temporal variance. Despite this physical difficulty, OceanEmbed v3 reduced thermocline error by up to 0.18°C vs CNN.

- Q3: Can deep learning replace physical numerical models like NEMO?**

Answer: No. OceanEmbed serves as an ultra-fast statistical reconstruction emulator (~35 ms) complementary to dynamical assimilation systems.

CONCLUDING SCIENTIFIC STATEMENT

Master Project Takeaway:

"OceanEmbed demonstrates the feasibility of reconstructing subsurface ocean temperature from surface satellite observations using a compact deep-learning architecture with independent observational checking."